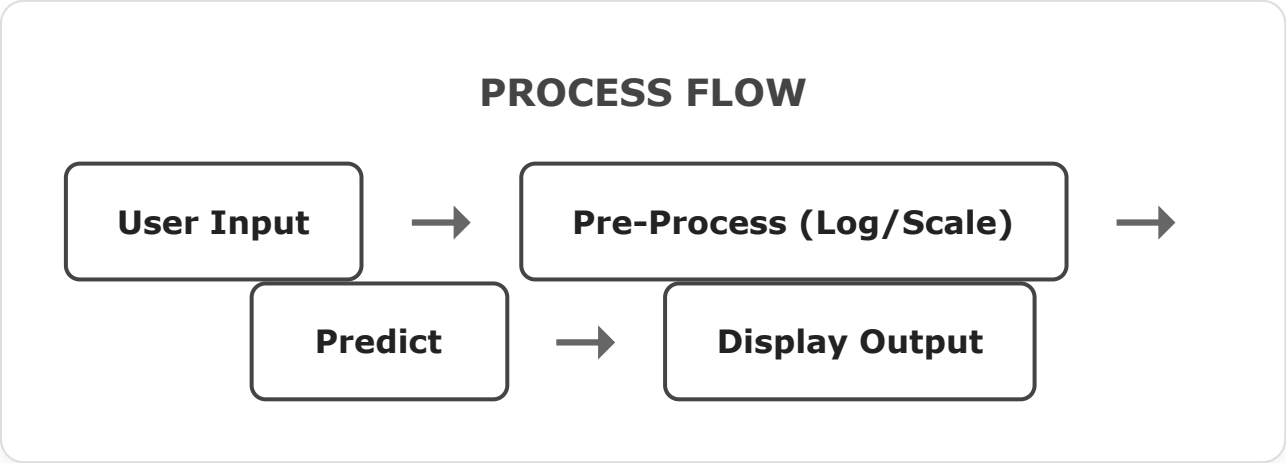
3. DEPLOYING WITH STREAMLIT

THE MISSION: "GOING LIVE"

You have trained the "Brain" (Model). But right now, it only lives in a Jupyter Notebook. Bank managers can't run code!

Your mission is to build a **User-Friendly Website** where a manager can simply click buttons to approve or reject loans. We will use a tool called **Streamlit**.

PHASE 1: THE WEB APP STRUCTURE



> 💡 **SIMPLE EXPLANATION:**
>
> *We will split the **app.py** code into two parts to make it easier to understand.*
> ***Part 1:*** *The Setup & Brain (Loading models and defining logic).*
> ***Part 2:*** *The Face (The Website Design).*

PART 1: THE SETUP & LOGIC

Create a file named `app.py` and copy this into the top.

PYTHON CODE

```python
import streamlit as st
import numpy as np
import joblib

# 1. LOAD THE BRAIN (Models)
# We use try-except to handle errors if files are missing
try:
    scaler = joblib.load("minmaxscaler.joblib")
    model = joblib.load("modellog.joblib")
except Exception as e:
    st.error(f"Error loading files: {e}")

# 2. DEFINE THE PREDICTION LOGIC
def prediction(Gender, Married, Dependents, Education, Self_Employed,
               Applicant_Income, Coapplicant_Income, LoanAmount,
               Loan_Amount_Term, Credit_History, Property_Area):

    # A. Convert Text to Numbers (0 or 1)
    # We use simple "if-else" logic, just like LabelEncoder did automatically
    gender_val = 1 if Gender == "Male" else 0
    married_val = 1 if Married == "Yes" else 0
    education_val = 0 if Education == "Graduate" else 1
    self_emp_val = 1 if Self_Employed == "Yes" else 0
    prop_val = 0 if Property_Area == "Rural" else 1 if Property_Area == "Semiurban" else 2

    # Handle Dependents (0, 1, 2, 3+)
    if Dependents == '3+': dep_val = 3
```

```
    else: dep_val = int(Dependents)

    # B. Feature Engineering (Total Income)
    total_income = Applicant_Income +
Coapplicant_Income

    # C. Log Transformation (Shrink huge numbers)
    # We add +1 to avoid log(0) errors
    income_log = np.log(total_income + 1)
    loan_log = np.log(LoanAmount + 1)
    term_log = np.log(Loan_Amount_Term + 1)

    # D. Prepare Input Array (ORDER MATTERS!)
    # Order: Gender, Married, Dependents, Education,
Self_Employed,
    #        ApplicantIncome, LoanAmount,
Loan_Amount_Term, Credit_History, Property_Area
    input_data = np.array([[gender_val, married_val,
dep_val, education_val,
                            self_emp_val, income_log,
loan_log, term_log,
                            Credit_History,
prop_val]])

    # E. Scale the Input (0 to 1)
    scaled_input = scaler.transform(input_data)

    # F. Predict
    return model.predict(scaled_input)[0]
```

PART 2: THE APP INTERFACE

Copy this below Part 1 in `app.py`.

PYTHON CODE

```python
def main():
    st.title("Welcome to Loan Application")
    st.header("Please enter your details:")

    # Create Dropdowns and Boxes for User Input
    Gender = st.selectbox("Gender", ("Male", "Female"))
    Married = st.selectbox("Married", ("Yes", "No"))
    Dependents = st.selectbox("Dependents", ("0", "1", "2", "3+"))
    Education = st.selectbox("Education", ("Graduate", "Not Graduate"))
    Self_Employed = st.selectbox("Self Employed", ("Yes", "No"))

    Applicant_Income = st.number_input("Applicant Income", min_value=0)
    Coapplicant_Income = st.number_input("Coapplicant Income", min_value=0)
    LoanAmount = st.number_input("Loan Amount", min_value=0)
    Loan_Amount_Term = st.number_input("Loan Term (Months)", min_value=0)
    Credit_History = st.selectbox("Credit History (1.0 = Good, 0.0 = Bad)", (1.0, 0.0))
    Property_Area = st.selectbox("Property Area", ("Rural", "Urban", "Semi Urban"))

    # THE BUTTON
    if st.button("Predict Loan Status"):
        result = prediction(Gender, Married, Dependents, Education, Self_Employed,
                            Applicant_Income, Coapplicant_Income, LoanAmount,
```

```
                                Loan_Amount_Term,
Credit_History, Property_Area)

        if result == 1:
            st.success("Congratulations! Your loan
application is APPROVED.")
        else:
            st.error("Sorry, your loan application is
REJECTED.")

if __name__ == "__main__":
    main()
```

💡 **SIMPLE EXPLANATION:**

Running the App: *Open your terminal and type:*

`streamlit run app.py`